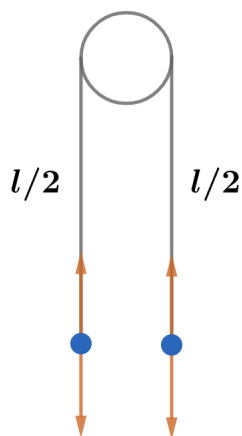


2020B F=ma Exam: Problem 10

Kevin S. Huang



The effective spring constant is $2k$ for a given mass since the spring length is effectively halved. Recall the period of a spring-mass system is given by

$$T = 2\pi\sqrt{\frac{m}{k}}$$

In our case,

$$T = 2\pi\sqrt{\frac{m}{2k}} = \pi\sqrt{\frac{2m}{k}}$$

so the answer is B.